From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2014

FULL TEST – IV

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-B**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (09 – 12) contains 4 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (13 – 18) contains 2 paragraphs. Based upon paragraph, 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+4 marks** for correct answer and **– 1 marks** for wrong answer.

2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 mark** will be awarded. There may be one or more than one correct choice. No marks will be given for any wrong match in any question. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
	=		0.0821 Lit atm $K^{-1}\ mol^{-1}$
	=		$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
	=		6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV	=		1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Single Correct Choice Type

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A longitudinal wave is represented by

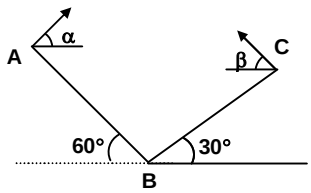
$$y = x_0 \text{ Sin } 2\pi\left(nt - \frac{x}{\lambda}\right)$$

The maximum particle velocity will be four times the wave velocity if

(A) $\lambda = \frac{\pi x_0}{4}$ (B) $\lambda = 2\pi x_0$

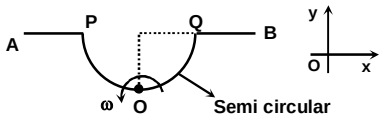
(C) $\lambda = \frac{\pi x_0}{2}$ (D) $\lambda = 4\pi x_0$

2. Two inclined planes AB and BC are at inclination of 60° and 30° as shown in the figure. The two projectiles of same mass are thrown simultaneously from A and C with speed 2 m/s and v_0 m/s respectively, they strike at B with same speed. If length of AB is $\frac{1}{\sqrt{3}}$ m and BC is 1 m, then find the value of v_0



(A) (1/2) m/s (B) 1 m/s
(C) 2 m/s (D) none of these

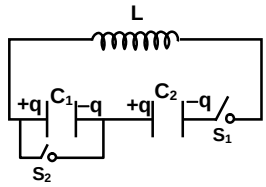
3. A conductor AB is rotated about O as shown in the figure such that its axis of rotation is parallel to the magnetic field $\vec{B} = B_0\hat{k}$ with uniform angular speed ω_0. The potential difference between points A and B is (given AP = QB)



(A) $5B_0\omega R^2$ (B) $-5B_0\omega R^2$
(C) zero (D) None

Space for Rough work

4. Two capacitors $C_1 = C_2 = \frac{1}{\pi^2}\times 10^{-2}$F and inductor L = 2×10^{-2} H are connected in series as shown in the figure. Initially charge on each capacitors are $4\sqrt{3}$ μC. At t = 0 switch S_1 is closed and t = 1/400 sec, switch S_2 is also closed. The maximum charge on capacitor C_2 during LC oscillation is



(A) 3 μC (B) 6 μC
(C) $3\sqrt{3}$ μC (D) $6\sqrt{3}$ μC

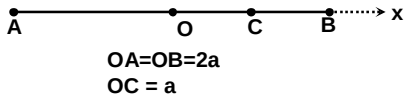
5. A light ray incident along vector $2\hat{i} + 4\hat{j} + \sqrt{5}\hat{k}$ strikes on the x-z plane from medium I of refractive index $\sqrt{3}$ and enters into medium II of refractive index is μ_2. The value of μ_2 for which the value of angle of refraction becomes 90°, is

(A) $\frac{4\sqrt{3}}{5}$ (B) $\frac{3\sqrt{3}}{5}$
(C) $\frac{2\sqrt{3}}{5}$ (D) $\frac{\sqrt{3}}{5}$

6. The energy of an electron in excited Hydrogen atom is −3.4 eV. Then according to Bohr's Theory the angular momentum of this electron in Js is

(A) 2.11×10^{-34} (B) 3×10^{-34}
(C) 2×10^{-34} (D) 0.5×10^{-34}

7. A particle of mass m is performing simple harmonic motion along line AB with amplitude 2a with centre of oscillation as O. At t = 0 particle is at point C (OC = a) and is moving towards B with velocity v = $a\sqrt{3}$ m/s. The equation of motion can be given by



(A) x = 2a (sin t + $\sqrt{3}$ cos t) (B) x = 2a ($\sqrt{3}$ sin t + cos t)
(C) x = a (sin t + $\sqrt{3}$ cos t) (D) x = a ($\sqrt{3}$ sin t + cos t)

8. Four identical samples of same polyatomic ideal gas undergo addition of same amount of heat under four different processes namely $PV^{1/2} = C$; $PV^{3/2} = C$; $P = C$ and $PV^2 = C$. The final internal energy will be maximum for the sample which has undergone

(A) $PV^{1/2} = C$ (B) $PV^{3/2} = C$
(C) $P = C$ (D) $PV^2 = C$

Space for Rough work

Multiple Correct Answer(s) Type

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

9. A parallel plate capacitor with plate area A and separation d has charge Q. A slab of dielectric constant k is inserted in space between the plates almost completely fills the space. If E_0, U_0 and C_0 be the electric field, potential energy and capacitance before inserting the slab, then

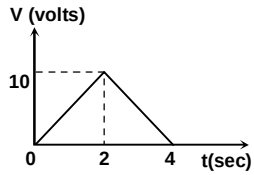
(A) the electric field after inserting the slab is $\frac{E_0}{k}$

(B) the capacitance after inserting the slab is $k\,C_0$

(C) the induced charge on the slab is $Q\left(1-\frac{1}{k}\right)$

(D) the energy stored in the capacitor becomes $\frac{U_0}{k}$, U_0 being the energy of the capacitor before inserting the slab

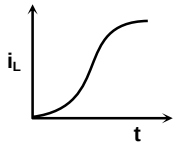
10. The potential difference across a 2H inductor as a function of time is shown in the figure. At time t = 0, current is zero. Choose the correct statement



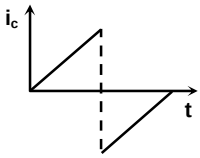
(A) Current at t = 2 sec is 5A

(B) Current at t = 2 sec is 10A

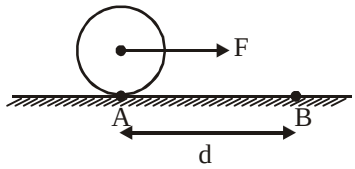
(C) Current vs time graph across inductor will be



(D) Current vs time graph across inductor will be



11. A uniform ring is connected to a light axle with light spokes so as to form a wheel. The wheel is placed on a horizontal surface with its plane vertical and a constant horizontal force F is applied to the axle. Surface AB is rough and surface right of B is smooth. The wheel does not slip when it moves from A to B. The wheel moves from A to B in time T.



(A) Rotational KE remains constant right of B

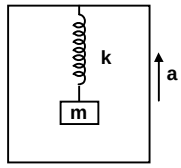
(B) Energy is dissipated from A to B

(C) Linear acceleration right of B is more than linear acceleration left of B

(D) The wheel undergoes constant non–zero acceleration to the right of B.

Space for Rough work

12. A spring mass system hanging from the ceiling of an elevator is at rest. The elevator starts accelerating upwards with acceleration 'a'. 'm' is the mass of block and 'k' is the spring constant of spring. Then Choose the correct alternative(s)



(A) Amplitude of oscillation is $\frac{ma}{k}$ (B) Amplitude of oscillation is $\frac{mg}{k}$

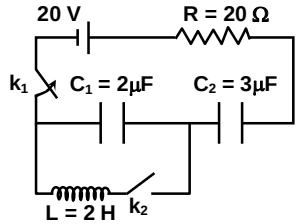
(C) Time period of oscillation is $2\pi\sqrt{\frac{m}{k}}$ (D) Time period of oscillations is $2\pi\sqrt{\frac{2m}{k}}$

Paragraph Type

This section contains 6 **multiple choice questions** relating to two paragraphs with **three questions on each paragraph**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

Paragraph for Questions 13 and 15

A circuit shown in the figure in which k_1 is closed and k_2 is open. Inductor L can be connected to capacitor C_1 by closing switch k_2 and opening k_1.



13. Let switch k_1 is closed and k_2 is opened for long time. The charge on capacitor C_2 will be
(A) 60 μC (B) 12 μC
(C) 24 μC (D) 8 μC

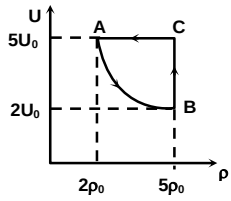
14. At t = 0, when capacitors are fully charged, switch k_1 is opened and switch k_2 is closed, so that inductor is connected in series with capacitor C_1. The maximum charge will appear on capacitor C_1 at time t is
(A) $\frac{\pi}{2000}$ sec (B) $\frac{\pi}{1000}$ sec
(C) $\frac{\pi}{500}$ sec (D) None

15. The maximum energy across inductor will be
(A) 0.144 mJ (B) 0.288 mJ
(C) 0.072 mJ (D) None of these

Space for Rough work

Paragraph for Questions 16 and 18

Figure shows the variation of the internal energy U with the density ρ of one mole of ideal monoatomic gas for a thermodynamics cycle ABCA. Here process AB is a part of rectangular hyperbola



16. The total amount of heat absorbed by the system for cyclic process is

(A) $\left(\frac{10}{3}\ln 2.5 - 2\right)U_0$ (B) $\left(\frac{10}{5}\ln 0.4 - 2\right)U_0$

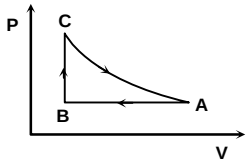
(C) $50U_0$ (D) None of these

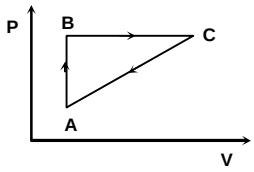
17. The work done in process AB is

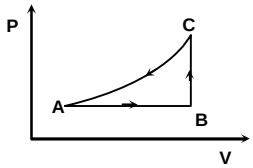
(A) $-U_0$ (B) $-2U_0$

(C) $-5U_0$ (D) none of these.

18. The P-V diagram of above process

(A) 

(B) 

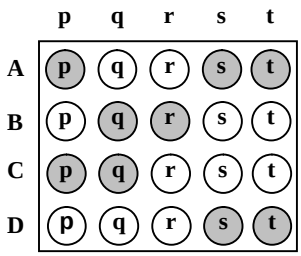
(C) 

(D) None of these

Space for Rough work

SECTION - B
Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*



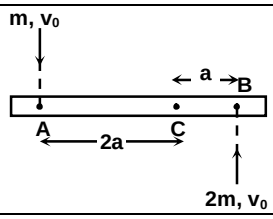
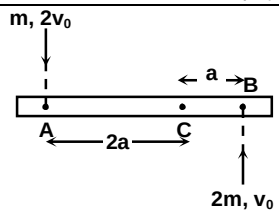
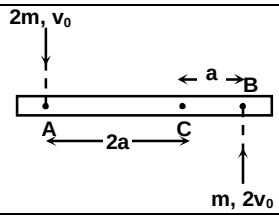
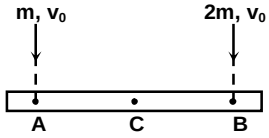
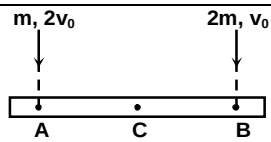
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. An electron in a hydrogen atom makes a transition $n_1 \rightarrow n_2$, where n_1 and n_2 are the principal quantum numbers of the two states. Assume Bohr model to be valid.

Column I		Column II	
(A)	The electron emits an energy of 2.55 eV	(p)	$n_1 = 2, n_2 = 1$
(B)	Time period of the electron in the initial state is eight time that in the final state.	(q)	$n_1 = 4, n_2 = 2$
(C)	Speed of electron become two times	(r)	$n_1 = 5, n_2 = 3$
(D)	Radius of orbit of electron becomes 4.77A°	(s)	$n_1 = 6, n_2 = 3$
		(t)	$n_1 = 8, n_2 = 4$

Space for Rough work

2. A rod of mass 6m and length 6a is kept at horizontal smooth surface. Two point masses of masses 2m and m respectively moving perpendicular to the rod, collide to the rod and stick to the rod after collision as shown in the column – II and stick to the rod. Point C represents centre of mass of the rod. AC = 2a and CB = a. v is the linear speed of the centre of mass of the system just after collision and ω is the angular speed of the system just after collision.

	Column – I		**Column – II**
(A)	$v = 0$	(p)	
(B)	$\omega = 0$	(q)	
(C)	$v \neq 0$	(r)	
(D)	$\omega \neq 0$	(s)	
		(t)	

Space for Rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Boron cannot form which of the following anions:

(A) BF_6^{3-} (B) BH_4^-

(C) $B(OH)_4^-$ (D) BO_2^-

2. Which of the following resonating structures of 1-methoxy-1,3-butadiene is most stable:

(A) $\overset{(-)}{C}H_2 - CH = CH - CH = \overset{\oplus}{O} - CH_3$ (B) $CH_2 = CH - \overset{(-)}{C}H - CH = \overset{\oplus}{O} - CH_3$

(C) $CH_2 = CH - \overset{\oplus}{C}H - \overset{(-)}{C}H - O - CH_3$ (D) $CH_2 = CH - \overset{(-)}{C}H - \overset{\oplus}{C}H - O - CH_3$

3. A mixture of two salts is not water soluble but dissolves completely in dilute HCl to form a colourless solution. The mixture could be

(A) $AgNO_3$ and KBr (B) $BaCO_3$ & ZnS

(C) $FeCl_3$ and $CaCO_3$ (D) $Mn(NO_3)_2 + MgSO_4$

4. The weight of a residue obtained by heating 2.76 g of silver carbonate is :

(A) 2.76 g (B) 2.98 g

(C) 2.16 g (D) 2.44 g

Rough Work

5. The brown ring complex $[Fe(H_2O)_5NO]SO_4$ has oxidation number of Fe
(A) +1 (B) +2
(C) +3 (D) +4

6. Black Jack is an ore of:
(A) Cr (B) Sn
(C) Zn (D) Ni

7. The correct order of decreasing acidic strength of oxides is:
(A) $Li_2O > BeO > B_2O_3 > CO_2 > NO_2$ (B) $N_2O_3 > CO_2 > B_2O_3 > BeO > Li_2O$
(C) $CO_2 > N_2O_3 > B_2O_3 > BeO > Li_2O$ (D) $B_2O_3 > CO_2 > N_2O_3 > Li_2O > BeO$

8. $C_5H_{13}N$ reacts with HNO_2 to give an optically active alcohol. The compound is:
(A) Pentan-1-amine (B) Pentan-2-amine
(C) N, N – dimethylpropan-2-amine (D) N – Methylbutan-2-amine

Rough Work

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

9. If P° and P_S are the V.P. of solvent and its solution respectively and N_1 and N_2 are the moles of solute and solvent then

(A) $(P° - P_S)/P° = N_1/(N_1 + N_2)$ (B) $(P° - P_s)/P_S = N_1/N_2$

(C) $(P° - P_S)/P° = N_1/N_2$ (D) None

10. Select the correct statements:

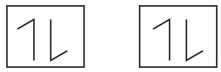
(A) For the reaction $H_{2(g)} + I_{2(g)} \underset{K_b}{\overset{K_f}{\rightleftharpoons}} 2HI\,(g)$; $\frac{d[H_2]}{dt} = K_b[HI]^2$ (Assuming the reactions to be elementary)

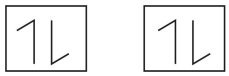
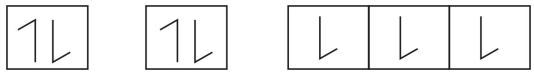
(B) At equilibrium net rate of change in conc. of H_2 is zero for $H_2 + I_2 \rightleftharpoons 2HI$

(C) The equilibrium constant for the reaction $H_{2(g)} + I_{2(g)} \underset{K_b}{\overset{K_f}{\rightleftharpoons}} 2HI$; is independent of temperature

(D) (A) is correct and (B) is wrong

11. Ground state of nitrogen atom can be represented by

(A)  (B) 

(C)  (D) 

12. Radioactive $\left(\overset{82}{Br}—\overset{82}{Br}\right)$ adds to 1-bromocyclohexene in CCl_4. The product is :

(A) 1, 1, 2 – tribromocyclohexane

(B) has radioactive bromine in vicinal position

(C) has radioactive bromine trans to each other

(D) has radioactive bromine cis to each other

Rough Work

Paragraph Type

This section contains 6 **multiple choice questions** relating to two paragraphs with **three questions on each paragraph**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

Paragraph for Question Nos. 13 to 15

A solid crystal is composed of X,Y and Z atoms, Z atoms (radius 200 pm) form cubic close packed structure, whereas X (radius 50 pm) and Y (radius 100 pm) occupy respective voids – Assume all the voids are occupied. Now answer the following questions.

13. If all the atoms along one body diagonal are removed, then the formula of the compound becomes :

(A) $X_8Y_4Z_5$ (B) $X_8Y_4Z_4$

(C) $X_6Y_3Z_3$ (D) $X_6Y_3Z_4$

14. Now in the crystal obtained in Q. No. 13 if all the atoms along another body diagonal are removed, then the formula of the compound becomes :

(A) $X_8Y_4Z_6$ (B) $X_8Y_6Z_5$

(C) $X_8Y_4Z_8$ (D) $X_8Y_6Z_7$

15. Which type of voids are occupied by X atoms:

(A) Tetrahedral void (B) Octahedral void

(C) Cubic void (D) All of these

Rough Work

Paragraph for Question Nos. 16 to 18

Buffer is a solution whose pH does not change significantly on addition of a small amount of acid or alkali or on dilution. Answer the following questions:

16. The amount of $(NH_4)_2SO_4$ to be added to 500 ml of 0.01 M NH_4OH solution (pKa for NH_4^+ is 9.26) prepare a buffer of pH 8.26 is :

(A) 0.05 mole (B) 0.025 mole

(C) 0.10 mole (D) 0.005 mole

17. The ratio of pH (approximately) of solution (I) containing 1 mole of CH_3COONa and 1 mole of HCl and solution (II) containing 1 mole of CH_3COONa and 1 mole of acetic acid in one litre is :

(A) 1 : 2 (B) 2 : 1

(C) 1 : 3 (D) 3 : 1

18. A weak acid HA after treatment with 12 ml of 0.1M strong base BOH has pH = 5. At the end point, the volume of same base required is 26.6 ml. K_a of acid is :

[Given : $\log\left(\frac{1.2}{1.46}\right) = -0.085$]

(A) 8.2×10^{-6} (B) 8.2×10^{-5}

(C) 8.2×10^{-4} (D) 8.2×10^{-7}

Rough Work

SECTION-B
(Matrix Type)

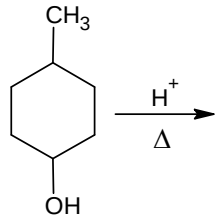
This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

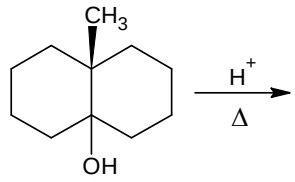
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

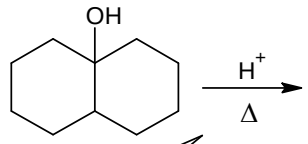
	p	q	r	s	t
A	●	○	○	●	●
B	○	●	●	○	○
C	●	●	○	○	○
D	○	○	○	●	●

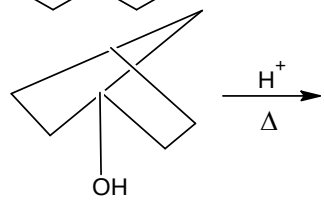
1. Match the Column – I with Column – II:

Column – I

(A) 

(B) 

(C) 

(D) 

Column – II

(p)

(q) Racemic mixture

(r) Major product consist of even number of α-hydrogen

(s) Will not undergo dehydration

(t) Major product consist of odd number of α-hydrogen

Bridged carbocation is an intermediate

2. Match the Column – I with Column – II:

Column – I(Reaction)	**Column – II(Products)**
(A) $[Co(NH_3)_4 (H_2O)_2] Cl_2$	(p) Geometrical isomers
(B) $[Pt (NH_3)_2 Cl_2]$	(q) Paramagnetic
(C) $[Co(H_2O)_5 Cl] Cl$	(r) Diamagnetic
(D) $[Ni(H_2O)_6] Cl_2$	(s) Metal ion with +2 oxidation state
	(t) On reaction with $AgNO_3$ forms 2 equivalents of AgCl

Rough Work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. OPQR is a square and M, N are the mid–points of the sides PQ and QR respectively. If the ratio of the areas of the square and the triangle OMN is k : 6, then $\frac{k}{4}$ is equal to
   (A) 2 (B) 4
   (C) 12 (D) 16

2. P is a point (a, b) in the first quadrant. If the two circles which pass through P and touch both the co–ordinate axes cut at right angles, then
   (A) $a^2 - 6ab + b^2 = 0$ (B) $a^2 + 2ab - b^2 = 0$
   (C) $a^2 - 4ab + b^2 = 0$ (D) $a^2 - 8ab + b^2 = 0$

3. If f(x) is a decreasing function then the set of values of 'k', for which the major axis of the ellipse $\frac{x^2}{f(k^2+2k+5)} + \frac{y^2}{f(k+11)} = 1$ is the x–axis, is
   (A) $k \in (-2, 3)$ (B) $k \in (-3, 2)$
   (C) $k \in (-\infty, -3) \cup (2, \infty)$ (D) $k \in (-\infty, -2) \cup (3, \infty)$

4. If $f(x) = 2\sin^2\theta + 4\cos(x+\theta)\sin x \sin\theta + \cos(2x+2\theta)$ then value of $f^2(x) + f^2\left(\frac{\pi}{4} - x\right)$ is
   (A) 0 (B) 1
   (C) –1 (D) x^2

5. If $\frac{a_0}{n+1} + \frac{a_1}{n} + \frac{a_2}{n-1} + ... + \frac{a_{n-1}}{2} + a_n = 0$, then the equation
   $a_0x^n + a_1x^{n-1} + a_2x^{n-2} + ... + a_{n-1}x + a_n = 0$ has
   (A) exactly one root in (0, 1) (B) at least one root in (0, 1)
   (C) no root in (0, 1) (D) at the most one root in (0, 1)

Space for rough work

6. If $f(x) = \sqrt{x}$, g(x) = $e^x - 1$, and $\int fog(x)dx = Afog(x) + B\tan^{-1}(fog(x)) + C$, then A + B is equal to

(A) 1 (B) 2
(C) 3 (D) none of these

7. Area bounded by the curve y = $\ell n\, x + \tan^{-1} x$ and x–axis from x = 1 to x = 2, is

(A) $\frac{5}{2}\ell n\ 2 - \frac{1}{2}\ell n\ 5 + 2\tan^{-1}2 - \frac{\pi}{3} - 1$ (B) $\frac{5}{2}\ell n\ 2 - \frac{1}{2}\ell n\ 5 + 2\tan^{-1}2 - \frac{\pi}{4} + 1$

(C) $\frac{5}{2}\ell n\ 2 - \frac{1}{2}\ell n\ 5 + 2\tan^{-1}2 + \frac{\pi}{4} - 1$ (D) $\frac{5}{2}\ell n\ 2 - \frac{1}{2}\ell n\ 5 + 2\tan^{-1}2 - \frac{\pi}{4} - 1$

8. If gradient of a curve at any point P(x, y) is $\frac{x+y+1}{2y+2x+1}$ and it is passes through origin, then curve is

(A) $2(x+3y) = \ell n\left|\frac{3x+3y+2}{2}\right|$ (B) $x + 3y = \ell n\left|\frac{3x+3y+2}{2}\right|$

(C) 3y + x = ℓ n (3x + 2y + 1) (D) $6y - 3x = \ell n\left|\frac{3x+3y+2}{2}\right|$

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

9. If $(4+\sqrt{15})^n = I + f$, where n is an odd natural number, I is an integer and 0 < f < 1, then

(A) I is natural number (B) I is an even integer

(C) (I + f)(1 – f) = 1 (D) 1 – f = $(4-\sqrt{15})^n$

10. A bag initially contains one red and two blue balls. An experiment consisting of selecting a ball at random, noting its colour and replacing it together with an additional ball of the same colour. If three such trials are made, then

(A) probability that atleast one blue ball is drawn is $\frac{9}{10}$

(B) probability that exactly one blue ball is drawn is $\frac{1}{5}$

(C) probability that all the drawn balls are red given that all the drawn balls are of same colour is $\frac{1}{5}$

(D) probability that atleast one red ball is drawn is $\frac{3}{5}$

Space for rough work

11. If z is a complex number satisfying $|z - i\,\mathrm{Re}(z)| = |z - \mathrm{im}(z)|$ then z lies on
(A) $y = x$ (B) $y = -x$
(C) $y = x + 1$ (D) $y = -x + 1$

12. In a four – dimensional space where unit vectors along axes are $\hat{i}$, $\hat{j}$, $\hat{k}$ and $\hat{\ell}$ and $\vec{a}_1$, $\vec{a}_2$, $\vec{a}_3$, $\vec{a}_4$ are four non–zero vectors such that no vector can be expressed as linear combination of others and $(\lambda-1)(\vec{a}_1-\vec{a}_2)+\mu(\vec{a}_2+\vec{a}_3)+\gamma(\vec{a}_3+\vec{a}_4-2\vec{a}_2)+\vec{a}_3+\delta\vec{a}_4=\vec{o}$ then
(A) $\lambda = 1$ (B) $\mu = -\frac{2}{3}$
(C) $\gamma = \frac{2}{3}$ (D) $\delta = \frac{1}{3}$

Paragraph Type

This section contains 6 **multiple choice questions** relating to two paragraphs with **three questions on each paragraph**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

Paragraph for Question Nos. 13 to 15

Read the following write up carefully and answer the following questions:
The complex slope of a line passing through two points represented by complex numbers z_1 and z_2 is defined by $\frac{z_2 - z_1}{z_2 - z_1}$ and we shall denote by ω. If z_0 is complex number and c is a real number, then $\bar{z}_0 z + z_0 \bar{z} + c = 0$ represents a straight line. Its complex slope is $-\frac{z_0}{\bar{z}_0}$.

Now consider two lines $\alpha\bar{z} + \bar{\alpha}z + i\beta = 0$ ….. (i)
and $a\bar{z} + \bar{a}z + b = 0$ ….. (ii)
where α, β and a, b are complex constants and let their complex slopes be denoted by ω_1 and ω_2 respectively

13. If the lines are inclined at an angle of 120° to each other, then
(A) $\omega_2\bar{\omega}_1 = \omega_1\bar{\omega}_1$ (B) $\omega_2\bar{\omega}_1^2 = \omega_1\bar{\omega}_2^2$
(C) $\omega_1^2 = \omega_2^2$ (D) $\omega_1 + 2\omega_2 = 0$

Space for rough work

14. Which of the following must be true
(A) a must be purely imaginary (B) β must be purely imaginary
(C) a must be real (D) b must be imaginary

15. If line (i) makes an angle of 45º with real axis, then $(1+i)\left(-\frac{2\alpha}{\bar{\alpha}}\right)$ is
(A) $2\sqrt{2}$ (B) $2\sqrt{2}$ i
(C) $2(1-i)$ (D) $-2(1+i)$

Paragraph for Question Nos. 16 to 18

Read the following write up carefully and answer the following questions:
Let $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ be two planes, where $d_1, d_2 > 0$. Then origin lies in acute angle if $a_1a_2 + b_1b_2 + c_1c_2 < 0$ and origin lies in obtuse angle if $a_1a_2 + b_1b_2 + c_1c_2 > 0$. Further point (x_1, y_1, z_1) and origin both lie either in acute angle or in obtuse angle, if $(a_1x_1 + b_1y_1 + c_1z_1 + d_1)(a_2x_1 + b_2y_1 + c_2z_1 + d_2) > 0$. One of (x_1, y_1, z_1) and origin lie in acute angle and the other in obtuse angle, if $(a_1x_1 + b_1y_1 + c_1z_1 + d_1)(a_2x_1 + b_2y_1 + c_2z_1 + d_2) < 0$

16. Given the planes $2x + 3y - 4z + 7 = 0$ and $x - 2y + 3z - 5 = 0$, if a point P is (1, –2, 3), then
(A) O and P both lie in acute angle between the planes
(B) O and P both lie in obtuse angle
(C) O lies in acute angle, P lies in obtuse angle
(D) O lies in obtuse angle, P lies an acute angle

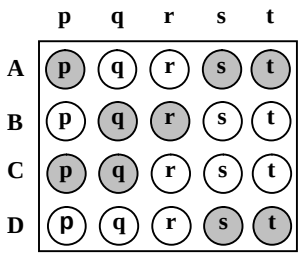
17. Given the planes $x + 2y - 3z + 5 = 0$ and $2x + y + 3z + 1 = 0$. If a point P is (2, –1, 2), then
(A) O and P both lie in acute angle between the planes
(B) O and P both lie in obtuse angle
(C) O lies in acute angle, P lies in obtuse angle
(D) O lies in obtuse angle, P lies an acute angle

18. Given the planes $x + 2y - 3z + 2 = 0$ and $x - 2y + 3z + 7 = 0$, if the point P is (1, 2, 2), then
(A) O and P both lie in acute angle between the planes
(B) O and P both lie in obtuse angle
(C) O lies in acute angle, P lies in obtuse angle
(D) O lies in obtuse angle, P lies an acute angle

Space for rough work

SECTION - B
Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*



If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

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1. Match the following Column–I with Column–II

Column – I		Column – II	
(A)	If $\cos A = \frac{\sin B}{2\sin C}$, then ΔABC is	(p)	isosceles
(B)	If $\frac{\cos A + 2\cos C}{\cos A + 2\cos B} = \frac{\sin B}{\sin C}$, then ΔABC may be	(q)	obtuse angle
(C)	If $\frac{2\cos A}{a} + \frac{\cos B}{b} + \frac{2\cos C}{c} = \frac{a}{bc} + \frac{b}{ca}$, then ΔABC is	(r)	right angle
(D)	If $\frac{a^2 - b^2}{a^2 + b^2} = \frac{\sin(A-B)}{\sin(A+B)}$, then ΔABC may be	(s)	acute angle
		(t)	equilateral

2. Match the following Column–I with Column–II

Column – I		Column – II	
(A)	If set of all possible values of k for which every solution of the inequation $x^2 - (3k - 1)x + 2k^2 - 3k - 2 \geq 0$ is also a solution of the inequation $x^2 - 1 \geq 0$ is $[\ell, m]$, then $\ell + m$ is equal to	(p)	3
(B)	If a, b, c and d are four positive real numbers such that abcd = 1 and minimum value of (1 + a)(1 + b)(1 + c)(1 + d) is 16λ, then $\lambda + 2$ is equal to	(q)	1
(C)	If solution set of the inequality $5^{x+2} > \left(\frac{1}{25}\right)^{1/x}$ is (ℓ, ∞), then ℓ is equal to	(r)	4
(D)	Let $f(x) = x^3 + 3x + 1$. If g(x) is the inverse function of f(x) and $g'(5) = 6\lambda$, then 4λ is equal to	(s)	2
		(t)	0

Space for rough work